ASSIGNMENT- 1

Q.1- Describe in detail the process of photosynthesis,highlighting the function of chloroplast.

Ans. Photosynthesis is the biochemical process by which plants, algae, and certain bacteria convert light energy into chemical energy (glucose) using water(H_2O) and carbon dioxide (CO_2), releasing oxygen (O_2) as a byproduct. Occurring within chloroplasts, this vital process uses chlorophyll to absorb sunlight, powering reactions that fuel plant growth and support nearly all life on Earth.

Key Components of Photosynthesis

- Reactants: Light energy, Water (H_2O), Carbon Dioxide (CO_2
- Products: Glucose ($C_6H_{12}O_6$), Oxygen (O_2).
- Location: Chloroplasts (specifically within the thylakoid membranes and stroma).
- Equation:
-

The Two-Stage Process

Photosynthesis takes place in two main stages:

1. Light-Dependent Reactions: Occurs in the thylakoid membranes. Chlorophyll absorbs sunlight, which splits water molecules (photolysis) to release oxygen, electrons, and protons. This stage converts solar energy into chemical energy, creating ATP and NADPH.
2. Calvin Cycle (Light-Independent Reactions): Occurs in the stroma. Using the ATP and NADPH produced in the first stage, the plant captures from the atmosphere to glucose.

FUNCTION OF CHLOROPLAST

Chloroplasts are specialized, double-membrane organelles found in plant cells and green algae that act as the primary "food producers" for plants. They are the site of photosynthesis, a complex process that converts light energy into chemical energy to fuel plant growth.

Key Functions of the Chloroplast

- Photosynthesis (Primary Function): Chloroplasts convert light energy, water (H_20), and carbon dioxide (CO_2) into oxygen(O_2) and chemical energy in the form of glucose ($C_6H_{12}O_6$) and ATP.

- Light-Dependent Reactions (in Thylakoids): Chlorophyll pigments within the thylakoid membranes trap sunlight, splitting water molecules (photolysis) to release oxygen and generate ATP and NADPH.
- Calvin Cycle/Dark Reactions (in Stroma): The stroma, the fluid-filled space surrounding the thylakoids, uses the ATP and NADPH produced in the light reactions to fix and produce sugar.
- Starch Storage: Excess glucose produced during photosynthesis is stored within the chloroplast as starch granules for future use.
- Biosynthesis Pathways: Chloroplasts are essential for the synthesis of various compounds, including fatty acids, lipids, amino acids, and plant hormones.
- Environmental Sensing and Stress Response: Chloroplasts act as sensors, detecting environmental changes like high light, drought, or extreme temperatures. They produce signaling molecules (like Reactive Oxygen Species) that initiate defense mechanisms and stress acclimation.

Functional Structure

- Thylakoid Membrane: The site of light-dependent reactions, where chlorophyll is embedded.
- Grana: Stacks of thylakoids that maximize surface area for light absorption.
- Stroma: A fluid-filled matrix where the Calvin cycle occurs and carbon is fixed.

Q.2 Analysis the different cell organelle and write their function.

Ans. Cell organelles are specialized, membrane-bound structures within eukaryotic cells that function like miniature organs to manage metabolism, energy production, and protein synthesis. Key organelles include the nucleus (genetic control), mitochondria (ATP energy generation), ribosomes (protein synthesis), endoplasmic reticulum (transport/synthesis), Golgi apparatus (packaging), and lysosomes (waste digestion).

Key Cell Organelles and Their Functions

- Nucleus: Often called the control center, it contains the cell's genetic material (DNA) and manages cell growth and reproduction.
- Mitochondria ("Powerhouse of the Cell"): Responsible for aerobic respiration, converting nutrients into ATP, which provides energy for cellular activities.
- Ribosomes: Tiny structures, found free or on the rough ER, that synthesize proteins from amino acids.
- Endoplasmic Reticulum (ER):
    - Rough ER: Studded with ribosomes, it synthesizes and modifies proteins.

    - Smooth ER: Lacks ribosomes, it synthesizes lipids, detoxifies harmful substances, and stores calcium ions.
- Golgi Apparatus/Complex: Functions as the cell's "post office," sorting, modifying, and packaging proteins and lipids into vesicles for delivery.
- Lysosomes: Known as "suicidal bags," these contain digestive enzymes that break down waste materials, foreign invaders, and old cell parts.
- Chloroplasts (Plant Cells only): Contain chlorophyll and conduct photosynthesis, converting sunlight into chemical energy (sugar).
- Vacuoles: Membrane-bound sacs, especially large in plant cells, used for storing water, nutrients, and waste products.
- Peroxisomes: Involved in lipid metabolism, breaking down fatty acids, and detoxifying toxic substances.
- Cytoskeleton: A network of protein fibers that provides structural support, shape, and facilitates movement within the cell.
- Centrioles (Animal Cells only): Cylindrical structures that organize microtubules during cell division.
- Cell Wall (Plant Cells only): A rigid outer layer that provides structural support and protection.

Analysis of Organelle Roles

These organelles ensure efficient cellular function through compartmentalization, allowing specialized biochemical processes to occur simultaneously without interference. While many are present in both plant and animal cells, specific structures like chloroplasts and cell walls are unique to plants, whereas lysosomes and centrioles are primarily found in animal cells.

Q.3 What are enzymes and outline different enzymes that are present in human body ?

Ans. Enzymes are specialized protein molecules that act as biological catalysts, accelerating essential chemical reactions in the human body without being consumed in the process. They function by binding to specific substrates at their "active site," lowering the activation energy required to convert these substances into products. Thousands of enzymes are essential for digestion, metabolism, and DNA replication.

Key Functions of Enzymes

- Catalysts: Speed up metabolic processes, respiration, and nerve function.
- Specificity: Each enzyme has a unique shape designed to bind with a specific substrate (lock-and-key mechanism).
- Conditions: They require specific temperatures and pH levels to function, often found in digestive juices or within cells.

Key Enzymes in the Human Body

Enzymes are generally categorized by the reactions they catalyze, with many ending in "-ase".

1. Digestive Enzymes (Digestive Tract, Pancreas, Mouth)

- Amylase: Breaks down carbohydrates and starches into sugars, found in saliva and pancreatic juice.
- Lipase: Breaks down fats into fatty acids and glycerol, found in pancreatic juice and fat-digesting organs.
- Protease: Breaks down proteins into amino acids, found in stomach acid and pancreatic juice (e.g., pepsin, trypsin).
- Lactase: Breaks down lactose (milk sugar).
- Sucrase: Breaks down sucrose (table sugar).
- Nuclease: Breaks down nucleic acids into nucleotides.

2. Metabolic and Cellular Enzymes

- ATP Synthase: Synthesizes Adenosine Triphosphate (ATP), the primary energy currency of cells, located in the mitochondria.
- Liver Enzymes: Include various cytochrome P450 enzymes that detoxify harmful substances and manage metabolism.
- Acetylcholinesterase: Breaks down the neurotransmitter acetylcholine, essential for nerve function.

3. DNA Replication and Repair Enzymes (Nucleus)

- DNA Polymerase: Responsible for synthesizing and replicating DNA strands during cell division.
- DNA Helicase: Unwinds the DNA double helix during replication.
- DNA Ligase: Joins DNA strands together.

4. Blood and Immune System Enzymes

- Clotting Enzymes (e.g., Thrombin): Essential for blood coagulation, acting in the circulatory system.
- Lysozyme: An enzyme found in tears and saliva that breaks down bacterial cell walls.

Classification of Enzymes

According to the International Union of Biochemistry and Molecular Biology (IUBMB), enzymes are classified into six major types based on their actions:

1. Oxidoreductases: Catalyze oxidation-reduction reactions.
2. Transferases: Transfer functional groups.
3. Hydrolases: Break bonds using water (hydrolysis).
4. Lyases: Remove groups without hydrolysis.
5. Isomerases: Rearrange molecular structures.
6. Ligases: Join molecules together.

Enzymes are vital for life, and their deficiency or dysfunction can lead to various diseases.